# order51 RESULTS: R1 tail bound and R2 H7 completion data

PREREG sha256 12898051d9283d562357b88225aaeac9a3081c8fea76d31f7c872703c7092393 (hashed before numerics). Runtime ~3 s.

## R1: derived and tested

Bound: $\|R\| \le B2 = \|(a)\| + \|N'^2\Sigma_{k\neq 0}|c'_k|\| + x\|N^2\Sigma_{k\neq 0}|c_k|\| \le B1$ (crude tail $(N^2-1)/6$). This is rigorous from $L_B$, $|G| \le 1$ and the unitarity of kM.

- R1a validity: PASS, $\|R\| \le B2 \le B1$ at all 12 points.
- **R1b non-vacuity: FAIL, as predicted.** $B2/\|S\|$ ranges from 3.37 ($x = 1$) to 29.8 ($x = 8$), while $\|R\|/\|S\|$ is 0.03–0.14.

Where it stalls: the triangle inequality discards the (a) ≈ −(b) cancellation that is the smallness. The tail term grows like $x\cdot N^2/6$ while $\|S\|$ grows like $x$. No bound that treats the $k \neq 0$ terms in absolute value can explain the near-law. Explaining it needs a phase-aware evaluation of $\Sigma_{k\neq 0}c_k G_{\beta+\lambda_k}$, which is the order46 obstruction.

## R2: derived data, one preregistered check FAILs as written

| | exact |
|---|---|
| c1 = $A^{-1}(1,0)^T$ (form ν1, ray (0,1)) | (−1/5, 1/5), Q(c1) = −1/10 |
| c2 = $A^{-1}(1,21)^T$ (form ν1+21ν2, ray (21,−1)) | (4, −1), Q(c2) = −17/2 |
| B(c1,c2) | **+4** (PREREG expected −4: hand-arithmetic sign error in c1) |

Both walls are negative vectors, and the completion constants are $\sqrt{1/-Q(c)}$ = √10 and √(2/17). But with the orientation fixed in PREREG, B(c1,c2) > 0, so the Zwegers same-component hypothesis FAILS as preregistered. With −c1 (form −ν1) it would hold (B = −4). Whether H7's sign function uses ν1 or −ν1 with c2 is fixed by H7's definition (the sign combination in the K± terms). That was not derived here, so I make no claim.

Stalled: forming A1 − $\Xi_{compl}$ requires the order-$t^{-1/2}$ coefficient of the erf-completion terms at rational x. That is not derivable from written sources here. No S-law test was run.

## Status

Category A: R1 inequality (valid but vacuous); exact c1, c2, Q values. Category B: none. v13: no. Ξ: not identified. E1–E3 OPEN.

Open next: (1) read off the sign orientation of the K± walls in H7 from the Part II definition and settle the Zwegers hypothesis; (2) derive the $t^{-1/2}$ asymptotic of $\Sigma(E(s_c\sqrt{y}) - \mathrm{sgn}\, s_c)\cdot$phase at rational x, then preregister the A1 − $\Xi_{compl}$ S-law test; (3) for smallness, a phase-aware (not absolute) treatment of the $k \neq 0$ $L_B$ terms.

# order51 NOTES

Disclosed: a sign error in the PREREG hand computation of B(c1,c2); the verdict stands as preregistered. R1 FAIL was predicted in advance. Naming: RG* only, no labels L1–L7; "35-component law vector". MAPS_CANDIDATES untouched; no Π/α1/fibre invented; no bridge claim. No push or sends.